Milestone 1: Data Loading and Cleaning

Project Overview:

The objective of this project is to analyze IT support ticket data to identify key performance trends, optimize resolution times, and improve overall service efficiency using data analytics. The project focuses on extracting meaningful insights from ticket data such as issue types, priority levels, and resolution performance.

The goal is to uncover patterns in customer requests, technical issues, and support team efficiency, enabling better decision-making and workflow optimization. This milestone establishes the foundation by preparing a clean and structured dataset for further analysis.

Key Objectives and KPIs:

The first milestone focuses on understanding the dataset and defining measurable performance indicators. The primary objectives include:

- **Loading and inspecting the dataset**
- **Understanding schema, data types, and data quality**
- **Cleaning and preparing the dataset for analysis**
- **Creating meaningful derived features**

The key performance indicators (KPIs) defined at this stage include:

- **Average Resolution Time**
- **First Response Time**
- **Ticket Volume by Type**
- **Priority Distribution**
- **Resolution Efficiency**

These KPIs will be used in later stages to evaluate IT support performance.

Dataset Overview:

The dataset is sourced from an internal IT support ticket system and contains approximately 8,000+ records spanning a two-year period. Each record represents a support ticket with multiple attributes.

The dataset includes the following types of information:

- **Ticket Details: Ticket ID, Type, Priority, Category**
- **Customer Information: Country**
- **Time Metrics: Date, Resolution Date, Resolution Time**
- **Clustering Attributes: Cluster ID, Cluster Name, Similarity Score**

This dataset provides a comprehensive view of support operations, enabling analysis of workload distribution, response efficiency, and recurring issues.

Data Loading and Exploration:

The dataset was loaded using pandas and explored to understand its structure and quality. Initial inspection involved:

- **Viewing sample records**
- **Checking data types of each column**
- **Identifying missing values**
- **Understanding categorical distributions**

This step helped identify inconsistencies such as incorrect data formats, missing values, and potential anomalies in the dataset.

Data Cleaning:

Data cleaning is a critical step to ensure accuracy and reliability of analysis. The following preprocessing steps were performed:

- **Handling missing values in key columns**
- **Removing duplicate records**

- **Standardizing categorical fields such as priority and ticket type**
- **Converting date-related columns into proper datetime format**
- **Validating consistency across related fields**

These steps ensured that the dataset became structured, consistent, and ready for further analysis.

Initial Data Analysis:

After cleaning, an initial analysis was performed to understand ticket distribution across different dimensions:

- **Distribution of tickets by type (e.g., request, incident, problem)**
- **Distribution by priority levels (low, medium, high, critical)**
- **Category-wise ticket frequency**

This analysis provided early insights into workload patterns and highlighted areas with higher ticket volumes or urgency levels.

Feature Engineering

To enhance analytical capabilities, new features were created from the existing dataset:

- **Resolution Duration: Calculated as the difference between resolution time and response time**
- **Priority Score: Numerical mapping of priority levels (Low = 1, Medium = 2, High = 3, Critical = 4)**

These features help in quantifying performance and simplifying analysis, especially for comparisons and modelling in later stages.

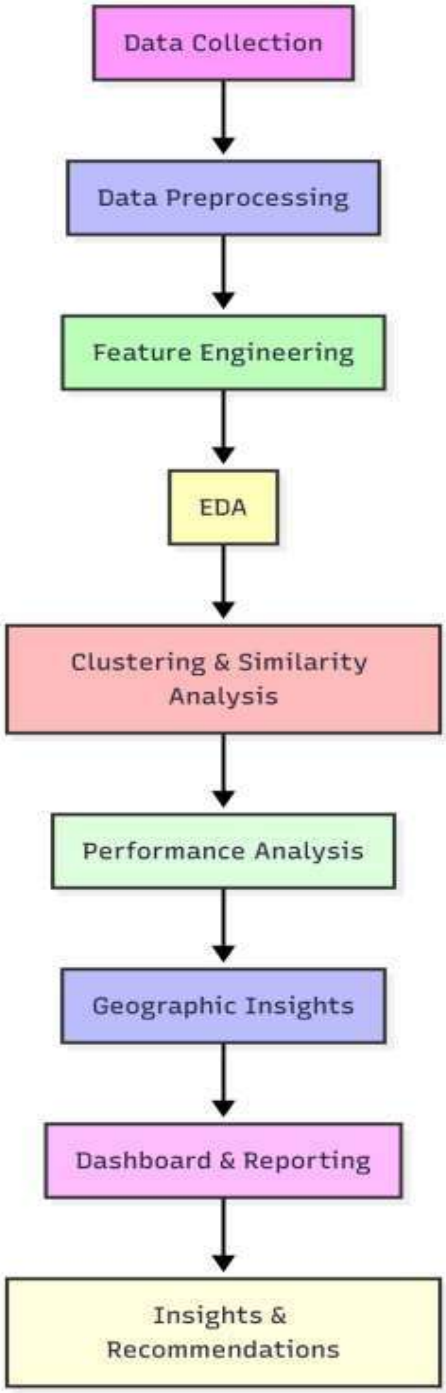
Project Workflow:

The workflow followed in this milestone reflects a structured data pipeline:

1. **Data Collection from IT support systems**
2. **Data Loading using pandas**
3. **Data Cleaning and preprocessing**

4. **Feature Engineering**
5. **Initial Data Analysis**

This systematic approach ensures that the dataset is reliable and ready for deeper analysis in subsequent



Deliverables:

The key outputs of this milestone include:

- **Cleaned and pre-processed dataset**
- **Feature-engineered dataset**
- **Summary of data transformations**
- **Initial insights on ticket distribution**

This milestone establishes a strong data foundation, ensuring that subsequent analysis, visualization, and modelling steps are built on accurate and well-structured data.

Conclusion:

Milestone 1 successfully prepares the dataset for advanced analytics by ensuring data quality, consistency, and usability. By defining KPIs, cleaning the data, and creating meaningful features, this stage lays the groundwork for extracting deeper insights in future milestones.

The processed dataset will be used in the next stages for exploratory data analysis, clustering, and performance optimization of IT support operations.